

Andres Perez

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WORK EXPERIENCE

University of Notre Dame

Notre Dame, IN

Student Researcher

Aug. 2024 – Present

- Assisting Prof. Chaoli Wang with research by data labeling image data and downloading GSV data of households, surveying heat dissipation in household.

Long Beach Unified School District

Carson, CA

Student Tutor

Sep. 2022 - May 2024

- Fostered an academic environment by leading peer-based pedagogy and promoting collaborative learning.
- Provided tutoring in subjects, including Geometry, Algebra 2, AP Pre-Calculus, AP Calculus AB & BC, AP Physics 1, AP Computer Science Principles, Computer Science Essentials, and Intro. To Japanese.

EDUCATION

University of Notre Dame

Notre Dame, IN

Bachelor of Science in Computer Engineering | Minor: Engineering Corporate Practice

Graduation Date: May 2028

PROJECTS

RFC Pi Simple Hat

Notre Dame, IN

Designer

- Designed a custom Arduino R3 hat PCB using KiCad to integrate and optimize Arduino and H-bridge circuits, improving system efficiency and reducing assembly complexity.
- Created and corrected the generated Bill of Materials and Component Placement List files from KiCad to export to JLCPCB for manufacturing.

Retrieval-Augmented Generation PDF Document Chat System

Long Beach, CA

Developer

- Integrated Python OpenAI's Chat Completion and Embedding APIs to generate context-aware responses based on semantic document understanding.
- Utilized Pinecone as a vector database to store and query embeddings for scalable and performant semantic search across multiple documents.
- Built a user-friendly interface using Streamlit, allowing real-time document upload, query input, and dynamic AI responses.

LEADERSHIP AND ACTIVITIES

IrishSat's Gravitational Orbital Attitude Thermal Lab

Notre Dame, IN

Electronics Lead

Aug. 2024 – Present

- Engineered a Python-based solution to streamline data acquisition from a magnetometer and Arduino, converting raw serial outputs into MATLAB-compatible formats for advanced analysis.
- Collaborated under expert mentorship to refine IrishSat's Helmholtz Cage design, leveraging KiCad for PCB design, MATLAB for data processing, and Raspberry Pi for system integration and control.

Long Beach Unified School District

Carson, CA

Ailurus Industries Electrical Team

Aug. 2023 – Jun. 2024

- Collaborated with a team of 30+ students to design, manufacture, and construct modular robots capable of reconfiguration for versatile tasks, including artifact retrieval.
- Enhanced UPenn's SMORES-EP cubic model by integrating an ESP32-UWB microcontroller, implementing a bevel gear design for differential drives. Applied NASA soldering standards in designing both physical body and electrical layouts, maintaining a 5-inch cube constraint for each module.

TECHNICAL AND LANGUAGE SKILLS

Programming: C, C++, Python, Git, MATLAB

Skills: Technical Writing, Electronics Soldering

Tools: Microsoft Office Suite, KiCad, Vim, SolidWorks, Fusion 360

Language: English (Native), Spanish (Bilingual)